



PRECISION METERING
FOR A CLEANER
ENERGY TOMORROW

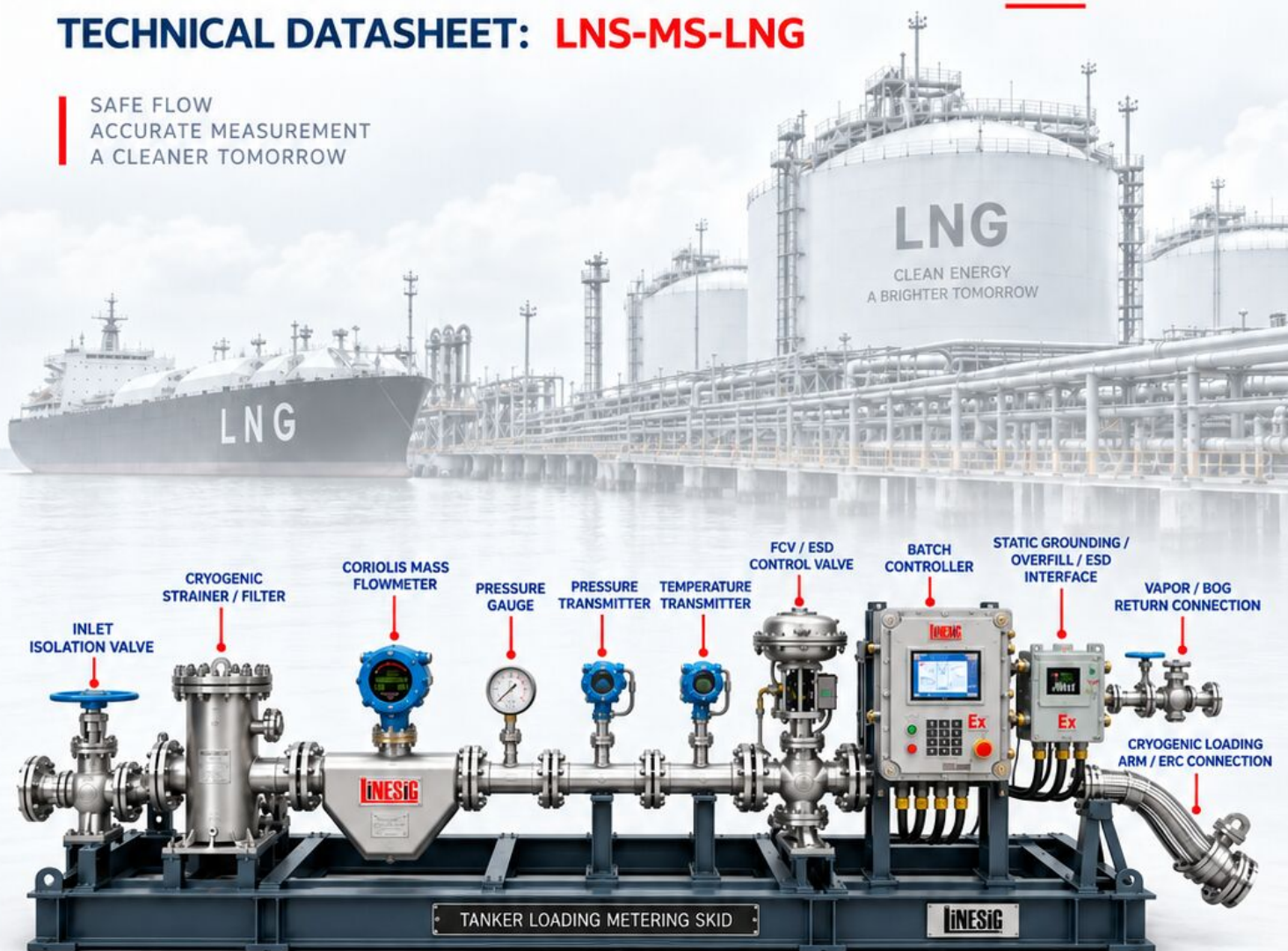
LNS-MS SERIES

Tanker Loading Metering Skids

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TECHNICAL DATASHEET: LNS-MS-LNG

SAFE FLOW
ACCURATE MEASUREMENT
A CLEANER TOMORROW



Product Model: **LNS-MS-LNG**

Application: Liquefied Natural Gas

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1. Product Overview

The **LINESIG LNS-MS-LNG** Series is a dedicated cryogenic skid-mounted tanker loading metering system for controlled and traceable transfer of liquefied natural gas. The architecture combines cryogenic Coriolis mass flow measurement, cryogenic pressure and temperature monitoring, batch/ESD control, controlled cooldown provisions, thermal relief for trapped liquid sections, cryogenic loading interfaces and vapour/BOG management according to the project loading philosophy.

Design philosophy:

LNS-MS-LNG is a cryogenic product configuration and is not an ambient-service skid with substituted materials. Final design requires project LNG composition, operating and design temperature/pressure, minimum design metal temperature, cooldown sequence, boil-off gas philosophy, loading-arm/ERC data, trapped-liquid thermal relief, thermal-movement analysis, insulation/vacuum-jacket philosophy and hazardous-area requirements.

Product Family Matrix:

Model	Primary Service	Preferred Metering	Special Design Considerations
LNS-MS-LH	Refined & light liquid hydrocarbons	Coriolis mass flowmeter	Custody/transaction measurement, loading arm interface, vapour return where required
LNS-MS-FO	Fuel oil, base oil & lube oil	Coriolis preferred; PD alternative where justified	Viscosity, heating/tracing, pressure drop, temperature compensation
LNS-MS-LPG	Liquefied petroleum gas	Coriolis mass flowmeter	Pressurized liquid service, vapour return/equalization, ESD, anti-flash hydraulic design
LNS-MS-LNG	Liquefied natural gas	Cryogenic Coriolis mass flowmeter	Cryogenic materials, cooldown, vapour/BOG interface, ESD and thermal contraction

2. LNS-MS-LNG - Liquefied Natural Gas Loading Skid

Design Aspect	LNS-MS-LNG Basis
Meter	Cryogenic Coriolis mass flowmeter
Filtration	Low-pressure-drop cryogenic strainer only where specifically required by project cleanliness / equipment-protection philosophy
Vapour / gas handling	Conventional atmospheric-liquid air eliminator not used; vapour/BOG return and liquid-phase stability are handled by the LNG process philosophy
Pressure / temperature	Cryogenic-rated PT and TT; temperature monitoring is essential for cooldown, transfer stability and protection
FCV / ESD	Fail-safe cryogenic control / batch / ESD valve arrangement with extended bonnet/stem design where required
BOG / vapour return	Dedicated BOG / vapour-return connection and isolation/control philosophy where required
Loading arrangement	Cryogenic loading arm or hose with ERC/breakaway philosophy
Batch controller	Explosion-proof local controller or TAS-integrated controller
Thermal movement	Supports, flexibility and insulation/vacuum-jacket philosophy project-specific
Thermal relief	Thermal relief valves on any section where cryogenic liquid can be blocked in between closed valves

Standard Process Sequence

Cryogenic Inlet Isolation / ESDV → Cryogenic Strainer where specifically required → Cryogenic Pressure / Temperature Measurement → Cryogenic Coriolis Mass Flowmeter → Cryogenic FCV / Batch / ESD Valve → Isolation → Cryogenic Loading Arm or Hose / ERC; controlled cooldown / recirculation and BOG / Vapour Return are provided as parallel connections where required. Blocked-in liquid sections are protected by thermal relief.

Functions

- Batch controller coordinates preset quantity, start/stop logic, cryogenic Coriolis totalisation and batch termination.
- Controlled cooldown establishes stable cryogenic conditions before normal loading and custody/batch measurement.
- ESD, gas-detection, loading-arm/ERC and vapour/BOG permissives are coordinated with the terminal LNG safety system.
- Thermal contraction, support loads, insulation / vacuum jacketing, external heat in-leak and trapped-liquid thermal relief are addressed during detailed engineering.



3. General Technical Specification (LNS-MS-LNG)

Parameter	Standard / Configurable Requirement
Product family	LNS-MS Series - tanker loading metering skids
Model	LNS-MS-LNG - Liquefied Natural Gas
Metering principle	Cryogenic Coriolis mass flowmeter
Flow range	Project-specific; meter and line size selected from minimum / normal / maximum loading rate
Line size	Project-specific
Design pressure	Project-specific
Design temperature	Project-specific cryogenic LNG service; operating temperature typically near -162 °C and design minimum established from project process / relief cases
Measurement outputs	Mass flow, mass total; density/temperature outputs where supported by selected cryogenic Coriolis meter
Meter accuracy	Vendor/model/project-specific; custody-transfer configuration where required
Pressure drop	Calculated for the complete cryogenic skid with adequate liquid-phase / anti-flash margin at minimum, normal and maximum flow
Process connection	ASME / EN / project flange standard
Piping material	Cryogenic-compatible stainless steel / project-specific
Insulation	Cryogenic insulation or vacuum-jacketed configuration according to allowable heat leak, cooldown and project philosophy
BOG / vapour return	Project-specific BOG / vapour-return connection, sized for cooldown and loading operating cases
Hazardous area	ATEX / IECEx / NEC / local requirement as specified by project
Electrical supply	Project-specific
Communication	Modbus RTU/TCP, Ethernet, pulse, frequency, analog/digital I/O as selected

Metering Technology Selection

- **Cryogenic Coriolis** is the standard **LNS-MS-LNG** metering approach for direct mass measurement at LNG temperature without requiring a conventional atmospheric-liquid air eliminator.
- Meter selection is coordinated with cooldown, liquid-phase stability, pressure drop, thermal contraction and insulation philosophy.
- The selected sensor, transmitter and seals must be specifically qualified for the project LNG temperature and hazardous-area requirements.

Selection Note:

Coriolis is preferred, not automatically mandatory. Final meter selection must account for flow range, viscosity, pressure drop, line size, custody requirements, product temperature and two-phase/flash behaviour.

4. Standard Scope of Supply

Item	Standard Scope / Description	Status
Skid base frame	Fabricated structural skid with lifting provision and equipment supports	Standard
Inlet isolation valve	Cryogenic manual isolation valve or actuated ESDV as specified	Standard / configurable
Strainer / filter	Low-pressure-drop cryogenic strainer where specifically required	Optional / project-specific
DP indication	Local DP gauge and/or DP transmitter only where a strainer/filter is fitted	Conditional
Vapour / BOG management	No atmospheric air eliminator; LNG vapour/BOG handled by cryogenic process design	Project-specific
Pressure gauge	Cryogenic-rated local pressure indication	Standard
Pressure transmitter	Cryogenic service pressure monitoring / batch or TAS interface	Standard / configurable
Temperature element / transmitter	Cryogenic temperature measurement for cooldown and transfer monitoring	Standard
Coriolis mass flowmeter	Primary metering device	Preferred standard
Flow control / batch valve	Cryogenic FCV / batch / ESD valve with fail-safe action as specified	Standard
Check valve	Cryogenic non-return valve where required by transfer hydraulic philosophy	Recommended
Explosion-proof batch controller	Local display, keypad, preset, totalisation and I/O	Standard for local batch control
Local ESD pushbutton	Emergency stop input to batch/terminal shutdown logic	Recommended
Grounding interface	Bonding / static-ground permissive for tanker loading where required	Recommended
Overfill interface	Independent overfill permissive/trip where applicable	Recommended
Junction / marshalling box	Field signal and power termination	As required
Loading arm / hose interface	Mechanical outlet connection to loading system	Interface provision
Cryogenic insulation / vacuum jacketing	Applied according to allowable heat leak, cooldown and project thermal design	Project-specific
BOG / vapour-return connection	Dedicated vapour return connection with cryogenic valves as required	Project-specific
Emergency release / ERC	Interface to loading arm or hose emergency-release system	Recommended / project-specific
Gas-detection / ESD interface	Hardwired or networked interface to terminal LNG safety system	Project-specific
Thermal relief valves	Relief of blocked-in LNG between isolation valves / equipment	Required where liquid can be trapped
Cooldown / recirculation connection	Controlled precooling / circulation branch as required by terminal operating philosophy	Project-specific

5- Explosion-Proof Batch Controller Functional Requirements

Function	Requirement
Preset quantity	Operator/TAS entry of target batch quantity
Live indication	Flow rate, batch delivered, target, product/status
Start/stop	Local controlled batch start and stop
Valve control	Open/ramp/close FCV or batch valve according to sequence
Meter input	Mass/volume pulses, frequency, digital or communication interface
Safety permissives	Cooldown complete, ESD, gas detection, route/pump, loading-arm/ERC, BOG/vapour-return, grounding and overfill permissives as applicable
Transaction data	Batch total, start/end time, alarms and operator/order reference
Communication	PLC/TAS/SCADA integration where required
Hazardous area	Certification selected to project classification

6. Instrumentation, Control & Safety Interface

Tag / Function	Typical Instrument / Device	Purpose
PI / PG	Cryogenic-rated pressure gauge	Local pressure indication
PT	Cryogenic pressure transmitter	Remote pressure monitoring / alarms
TT / TE	Cryogenic temperature transmitter / element	Cooldown, transfer monitoring and protection
GD	Gas-detection interface	LNG vapour detection / shutdown integration
CMF	Coriolis mass flowmeter	Mass flow, batch total and supported diagnostics
FCV	Cryogenic flow control / batch / ESD valve	Loading-rate control and batch termination
BC	Explosion-proof batch controller	Preset, totalisation, sequence and I/O
ESD-PB	Emergency stop pushbutton	Immediate loading stop command
GS	Static grounding monitor/interface	Ground permissive before and during transfer where applicable
OFP	Overfill protection interface	Independent overfill permissive/trip where applicable
LS/PS	Package switches as required	Interlocks / local package protection
TRV	Cryogenic thermal relief valve	Protects blocked-in liquid sections from thermal expansion

Minimum Batch Permissive - Typical

- Valid product / order / batch selected.
- Cryogenic route and source system ready.
- Cooldown sequence complete or approved loading condition established.
- Meter, batch controller and cryogenic valve chain healthy.
- BOG / vapour-return system ready where required.
- Loading-arm/hose and ERC interface healthy.
- No active ESD, gas-detection trip or critical skid fault.

Trip / Shutdown Actions - Typical

- ESD command: close batch/ESD valve and stop transfer-pump demand according to project cause-and-effect.
- Grounding loss during loading: controlled or immediate shutdown according to project safety philosophy where grounding is applicable.
- Overfill trip: stop batch and close transfer valve(s) where overfill protection is provided.
- Meter or controller critical fault: inhibit automatic batch completion until the approved safe fallback is defined.
- High/low pressure or abnormal process condition: alarm and trip where required by the approved design.
- Gas-detection or ERC trip: initiate the project-defined LNG emergency shutdown sequence.

7. Materials, Options & Ordering Data

Component	Typical Material / Option
Skid frame	Painted carbon steel / stainless steel option; thermal interfaces engineered as required
Process piping	Cryogenic-compatible austenitic stainless steel (e.g., 304L / 316L where specified) / project-specific
Meter wetted parts	Cryogenic-compatible materials selected with meter vendor
Strainer body	Cryogenic-compatible stainless steel where fitted
Valves	Cryogenic construction with extended bonnet/stem and qualified seat/seal materials where required
Gaskets / seals	Cryogenic-qualified, LNG-compatible materials selected for minimum design temperature
Instrument tubing	Cryogenic-compatible stainless steel with suitable supports and thermal allowances
Insulation / vacuum jacket	Project-specific thermal design

Available Options

- Vacuum-jacketed or insulated process piping, selected from the project heat-leak and cooldown philosophy.
- BOG / vapour-return package.
- Cryogenic loading-arm or hose ERC/breakaway interface.
- Additional PT/TT points for cooldown and thermal monitoring; thermal relief valves for blocked-in liquid sections.
- Proving/calibration connections where project metering philosophy permits.
- TAS / PLC / SCADA integration, remote I/O and historian interface.

8. Ordering Code - Proposed

LNS-MS / LNG / SIZE / METER / CONTROL / LOADING / OPTIONS

Field	Examples	Meaning
SERVICE	LNG	Liquefied Natural Gas variant
SIZE	Project DN / NPS	Nominal cryogenic skid process line size
METER	CMF-C	Cryogenic Coriolis mass flowmeter
CONTROL	EX-BC / TAS	Explosion-proof batch controller or TAS-integrated control
LOADING	CRYO-ARM / HOSE	Cryogenic loading interface
BOG	V	BOG / vapour-return interface
ESD / ERC	E / R	Cryogenic ESD / emergency release option
INSULATION	I / VJ	Insulated / vacuum-jacketed option

9. Project Data Required for Final Selection

Required Input	Project Entry
LNG composition	TBD
Minimum / normal / maximum liquid flow	TBD
Batch size / tanker capacity	TBD
Design / operating pressure	TBD
Design / operating temperature	TBD
Density / saturation condition	TBD
Cooldown procedure / rate	TBD
BOG / vapour-return flow and pressure	TBD
Line size / flange rating	TBD
Hazardous-area classification	TBD
Required metering accuracy / custody status	TBD
Cryogenic loading arm/hose / ERC interface	TBD
Insulation / vacuum-jacket requirement	TBD
Power supply / control interface	TBD
Material / thermal-stress requirements	TBD
Minimum design metal temperature / relief cases	TBD
Thermal relief / trapped-liquid philosophy	TBD

10. Document Notes

This technical datasheet is limited to the **LNS-MS-LNG** product configuration for Liquefied Natural Gas. Values identified as project-specific or TBD shall be completed on the project datasheet before final equipment selection, sizing, certification or manufacture.

